Deep Learning-Based Monocular Visual Odometry for Autonomous Spatial Mapping

Interim report on

Deep Learning Project

[ICT-4442]

Submitted By

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PART B : INTERIM REPORT

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Title of the Project (confirmed from Synopsis): Deep Learning-Based Monocular Visual Odometry for Autonomous Spatial Mapping

GitHub Repository Link: https://github.com/VaibhavBansal05/Deep_Monocular_VO

1. Literature Review

Nine papers, summarised and compared. Full per-paper summaries are maintained in the repository at src/reports/literature/.

Paper (Author, Year)	Method	Dataset	Key Result	Relevance to Project
Cabon et al., 2020 — Virtual KITTI 2	Unity-rendered photo-realistic clones of 5 KITTI tracking sequences; 10 variants each (weather, camera rotation); RGB, depth, flow, segmentation and 6-DoF poses	vKITTI 2 (synthetic) vs real KITTI	Re-ran depth, pose and segmentation baselines on both; synthetic data is a usable proxy, but a measurable domain gap remains	Our dataset. 100% dense depth (vs ~5% LiDAR coverage) and exact poses
Wang et al., 2017 — DeepVO (ICRA)	CNN (FlowNet-pretrained) to 2-layer LSTM to 6-DoF pose; fully end-to-end, no classical pipeline module retained	KITTI odometry	Competitive with classical methods; learns absolute scale implicitly, which monocular geometry cannot recover unaided	Model 2's blueprint, including the rotation loss weighting
Zhou et al., 2017 — SfMLearner (CVPR)	DepthNet + PoseNet trained only by photometric view-synthesis loss; learned explainability mask for non-rigid regions	KITTI odometry / Eigen split	ATE 0.021+/-0.017 (Seq 09) and 0.020+/-0.015 (Seq 10) over 5-frame snippets; beats ORB-SLAM (short) 0.064, behind ORB-SLAM (full) 0.014/0.012	Model 3's blueprint; pose is learnable with no labels at all
Godard et al., 2019 — Monodepth2 (ICCV)	Three targeted fixes: minimum-over-reference reprojection loss, full-resolution multi-scale sampling, and auto-masking of static pixels	KITTI (Eigen split)	Abs Rel 0.115 monocular at 640x192 (0.132 without ImageNet pretraining); ablations isolate each component	Model 3's loss design, item for item; also a template for our ablations
Zhao et al., 2019 — LKN-VO (Robot. Auton. Syst.)	Learning Kalman Network: deep observation and transition models fused by the Kalman update instead of an RNN hidden state	KITTI	Learns absolute scale without geometric constraints; filter parameters learned rather than hand-designed	The principled alternative to Model 2's recurrence
Zhi et al., 2022 — Multi-Scale Unsupervised MVO (Sensors)	Densely-linked atrous convolutions to widen the receptive field without downsampling, plus non-local self-attention for long-range dependencies	KITTI	Competitive with unsupervised SOTA; state-of-the-art rotation estimation, with gains concentrated in rotating scenes	Closest published analogue to Model 4; source of its central hypothesis
Albukhari et al., 2025 — MINI-DROID-SLAM (Sensors)	Replaces DROID-SLAM's ConvGRU update operator with a Mini-GRU; retains dense depth and bundle adjustment	TartanAir	ATE comparable to DROID-SLAM at reduced training time and computational cost, trainable on a single GPU	Evidence for Model 2's GRU-vs-LSTM ablation
Xie et al., 2025 — CoProU-VO (GCPR)	Projects reference-frame uncertainty into the target frame and combines it with the target's own under one probabilistic mask; ViT backbone	KITTI, nuScenes	Outperforms prior unsupervised two-frame end-to-end methods, with the largest margin on highway scenes where others fail	Most recent work in our family; natural extension to Model 3
Messikommer et al., 2024 — RL Meets VO (ECCV)	An RL agent inside a classical VO pipeline selecting keyframes and grid size online; reward from pose error and runtime	Public VO benchmarks	Improves accuracy and robustness while removing weeks of expert heuristic tuning	Shows learning can augment rather than replace classical VO; framing for Model 1

Synthesis.

The field has moved from replacing the geometric pipeline to repairing it. DeepVO and SfMLearner argued the whole pipeline could be learned end-to-end; the strongest later work is narrower, with Monodepth2 winning on three specific loss fixes and RL-VO leaving a classical pipeline intact while learning only its heuristics. This is why our four models deliberately span architecture families.

Supervision level and scale handling are entangled. Supervised models learn absolute scale from pose labels; unsupervised models recover structure only up to an unknown global scale. A comparison that ignores this flatters the supervised side, so our protocol reports ATE both with scale solved (sim3) and with scale fixed (SE3).

Attention is the least-settled axis and the gap this project addresses. Only two of the nine papers use it, and neither isolates it: Zhi et al. introduce atrous convolutions and non-local attention together, and CoProU-VO changes the backbone to a ViT. Model 4's ablation holds backbone, data, schedule and seed fixed and varies only the attention module, measuring something the literature leaves confounded.

None of the nine uses Virtual KITTI 2 as its primary benchmark. Our results are therefore not directly comparable to published KITTI numbers and are reported as an optimistic upper bound rather than as parity.

2. Dataset Acquisition & Preprocessing

Acquisition.

Virtual KITTI 2 is distributed as monolithic archives (~7.2 GB RGB, ~8 GB depth). scripts/prepare_data.py streams the archives and extracts only the scenes, variations and camera required by the split, yielding a 1.4 GB working subset of 8,642 frames across 18 sequences. Frame counts were verified against the declared scene lengths (Scene01 447, Scene02 233, Scene06 270, Scene18 339, Scene20 837).

Preprocessing pipeline.

- Resize. RGB frames are bilinearly resized from 1242x375 to 128x416.
- Intrinsics rescaling. K is scaled by the same factors as the image (fx, cx by width; fy, cy by height). This is required for the photometric reprojection in Model 3; an unscaled K produces a systematic reprojection error the loss cannot correct.
- Normalization. ImageNet mean/std, matching the pretrained ResNet-18 backbones.
- Temporal stacking. Two frames for the pairwise models (Models 1 and 4) and five frames for the sequence models (Models 2 and 3).
- Pose conversion. extrinsic.txt supplies world-to-camera matrices E. Relative pose is computed as T(t to t+1) = E(t+1) * inverse(E(t)), and absolute camera-to-world as inverse(E). Camera 0 rows are filtered explicitly, since both cameras are interleaved in the file.
- Augmentation. Colour jitter applied identically across all frames of a sequence, plus horizontal flip with the corresponding pose mirroring.

Split.

The split is by scene, so no geometric content is shared between train and test. Train: Scene01, Scene06, Scene18 under clone, morning, overcast and sunset (4,212 samples). Validation: Scene02/clone (232). Test: Scene20/clone (836). The rain, fog, 15-degree and 30-degree variations are held out entirely for a robustness ablation at the Final stage. The split is defined once in mvo/common/data/splits.py and is identical for all four models; unit tests assert that no scene appears in two splits.

Verification.

Integrating the ground-truth relative poses through the full pipeline reproduces the ground-truth trajectory as an exact rigid transform (residual 7e-14 m when the true initial pose is applied), confirming the pose conventions are consistent end to end. The extrinsic files store roughly seven significant digits, so their rotation matrices are orthonormal only to ~5e-7; chaining 837 of them accumulates to ~2e-3 m. This is the precision floor of the dataset and no model can be meaningfully scored below it.

3. Models Implemented So Far

Model	Owner (Member)	Status	Preliminary Metric	Notes
Model 4 — CNN + CBAM/SE attention	Vaibhav Krishna Bansal	Baseline trained (attention = none); CBAM/SE in progress	Best val loss 0.02543 at epoch 12 of 21; 12.50 M parameters. ATE / RPE: **<<pending evaluation>>**	ResNet-18 with conv1 inflated to 6 channels for the stacked pair. Validation loss minimised at epoch 12 and rose thereafter while training loss continued to fall.
Model 2 — CNN + GRU/LSTM	Abhishek Laxman Jadhav	Implemented; training in progress	**<<pending — training running>>**	15.39 M parameters, sequence length 5, 2-layer LSTM over a shared ResNet-18 encoder. Supports rnn = lstm | gru | none for ablation.
Model 1 — Classical ORB/LK + MLP	Pranav Joshi	Not started; dataset and preprocessing pipeline delivered	n/a	Feature extraction is deterministic and will be precomputed, so training is CPU-feasible.
Model 3 — Self-supervised PoseNet/DepthNet	Suryansh Bajaj	Not started; metrics and evaluation harness delivered	n/a	Longest pole; depends on the projection utilities, which are complete and tested.

Shared infrastructure.

All four models subclass a common BaseVOModel contract and are trained by a single engine and scored by a single evaluation script, so the comparison protocol cannot drift between members. The repository currently holds 60 automated tests covering the SE(3) utilities, projection, dataset contract, metrics and model interfaces, run in continuous integration on every pull request.

Rotation loss weighting.

The rotation weight k in the pose-regression loss was measured rather than taken from the literature. On the training split E[t^2] = 2.00e-1 against E[r^2] = 3.41e-5, a ratio of approximately 5869:1. DeepVO's published k = 100 leaves the rotation term about 59 times under-weighted at this scale, allowing the model to ignore rotation and still reduce the loss. Models 2 and 4 therefore use k = 5900, reproducible via scripts/measure_pose_stats.py.

4. Individual Contribution Log (to date)

Cross-checked against GitHub commit history (git shortlog -sne) on the main branch.

Member Name	Reg. No.	Task(s) Completed	Signature
Vaibhav Krishna Bansal	230911546	Repository scaffold and shared contracts (pose convention, batch schema, split, metrics schema); training engine; model registry; CI; Model 4 backbone and pose decoder; measured rotation-weight correction for Models 2 and 4; dataset acquisition and distribution; Colab and Kaggle training notebooks; literature review (3 papers). 33 commits.	
Pranav Joshi	230953248	vKITTI 2 subset downloader with streaming extraction; image transforms and intrinsics rescaling; VKitti2 dataset implementing the shared batch contract; depth loading; dataset unit tests. 5 commits.	
Abhishek Laxman Jadhav	230953360	SE(3) and SO(3) utilities (Rodrigues exponential and logarithm with small-angle stability, analytic inverse, relative/absolute trajectory chaining); perspective projection and inverse warping; Model 2 CNN+RNN architecture; geometry and projection test suites. 7 commits.	
Suryansh Bajaj	230953250	Umeyama similarity alignment with reflection guard; ATE with separate sim(3) and SE(3) reporting; RPE and KITTI-style relative drift; monocular depth metrics with median scaling; TUM trajectory I/O for cross-validation against evo; comparison-table generator; metrics test suite. 9 commits.	

5. Risk / Plan for Remaining Work

Remaining models.

Model 1 (classical + MLP) and Model 3 (self-supervised PoseNet/DepthNet) are not yet implemented. The shared infrastructure both depend on is complete and tested: Model 1 requires only the feature extractor and regressor, and Model 3's projection and warping utilities are already merged and unit-tested.

Risk	Assessment and mitigation
Model 3 fails to converge	High impact, moderate likelihood. Photometric self-supervision is the most fragile objective of the four. Mitigation: start immediately, keep the ImageNet-pretrained encoder (non-negotiable at our dataset size), and fall back to fewer scales or lower resolution. Monodepth2's reported 0.132-vs-0.115 gap without pretraining bounds the expected cost.
Overfitting on a small training set	Confirmed, not hypothetical. Model 4's validation loss minimised at epoch 12 and rose thereafter. The cause is identified: the augmentation RNG is seeded per sample index, so each sample receives identical jitter every epoch and augmentation is effectively inactive. A fix is scheduled; both trained models will be re-run afterwards.
GPU availability	Realised. The free Colab allowance was exhausted mid-run. Mitigation already in place: training migrated to Kaggle (approximately 30 h/week, headless execution up to 12 h), with local CPU training as a fallback at roughly 9 minutes per epoch for Model 4.
Unequal training budgets	Models trained for different numbers of epochs are not strictly comparable. For the Final submission all four will be re-run under one budget on the same hardware; the Interim figures are explicitly preliminary.
Uneven member availability	Two of four members were unavailable during the week before this submission. Tracked weekly via git shortlog so that workload can be reassigned early rather than at the deadline.

Timeline to Final submission.

Window	Planned work
25 Sep - 30 Sep	Fix the augmentation defect; implement Model 1 (Pranav) and begin Model 3 (Suryansh); complete Model 4's CBAM/SE variants (Vaibhav).
1 Oct - 12 Oct	Model 3 trained and evaluated. All four models re-run under an identical epoch budget on identical hardware.
13 Oct - 20 Oct	Hyperparameter tuning; ablations (attention on/off, LSTM/GRU/none); robustness evaluation on the held-out rain, fog and rotated variations; unified comparison table and error analysis.
21 Oct - 26 Oct	Final report Part C, one methodology subsection per model owner; slides; plagiarism check; viva rehearsal.
26 Oct - 31 Oct	Presentation and final submission with repository link.